

Combinatorics – 4

Graphs

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Theoretical minimum. Everything needed is proved here; a course in graph theory is not presupposed.

Definitions. A *graph* $G = (V, E)$ is a finite set of vertices with a set of unordered pairs as edges; $\deg v$ is the number of edges at v . A *path*, *cycle*, *connected*, *component*, *tree* (connected without cycles), *forest*, *bipartite* (the vertices split into two parts with all edges between them), *complete* K_n , *complete bipartite* $K_{m,n}$ have their usual meanings. A *tournament* is an orientation of K_n .

Theorem 1 (handshake and trees). $\sum_v \deg v = 2|E|$, so the number of vertices of odd degree is even. For a graph on n vertices the following are equivalent: it is a tree; it is connected with $n - 1$ edges; it is acyclic with $n - 1$ edges; between any two vertices there is exactly one path. Every tree with $n \geq 2$ vertices has at least two leaves, and the number of labelled trees on n vertices is n^{n-2} (Cayley; the Prüfer code is an explicit bijection with the words of length $n - 2$ over an alphabet of size n).

Theorem 2 (bipartiteness and Euler). A graph is bipartite if and only if it contains no cycle of odd length. A connected graph has a closed walk using every edge exactly once if and only if every degree is even (Euler), and an open one if and only if exactly two degrees are odd.

Theorem 3 (Hall's marriage theorem). A bipartite graph with parts A and B has a matching saturating A if and only if $|N(S)| \geq |S|$ for every $S \subseteq A$. *Proof* by induction on $|A|$: if every non-empty proper S has $|N(S)| > |S|$, match any vertex arbitrarily and apply induction; otherwise some S has $|N(S)| = |S|$, and the two induced subproblems again satisfy the condition. *Corollaries*: a regular bipartite graph has a perfect matching;

a doubly stochastic matrix is a convex combination of permutation matrices (Linear algebra – 11).

Theorem 4 (planarity). For a connected plane graph $v - e + f = 2$. Hence $e \leq 3v - 6$ for $v \geq 3$, and $e \leq 2v - 4$ if there is no triangle; consequently K_5 and $K_{3,3}$ are not planar, every planar graph has a vertex of degree at most 5, and by induction every planar graph is 6-colourable.

Fact 1 (the adjacency matrix). The number of walks of length k from u to v is $(A^k)_{uv}$; the largest eigenvalue satisfies $\bar{d} \leq \lambda_1 \leq \Delta$; the spectrum is symmetric about zero exactly for bipartite graphs. The Perron–Frobenius background is in Linear algebra – 11, the linear-algebraic method in Linear algebra – 6.

1. Prove the handshake lemma, the four characterisations of a tree, and Cayley’s formula by the Prüfer code. Then prove that a connected graph with n vertices and n edges has exactly one cycle, and count the labelled graphs of this shape.
2. Prove Hall’s theorem and deduce: a k -regular bipartite graph decomposes into k perfect matchings; a bipartite graph with parts of size n and minimum degree at least $n/2$ has a perfect matching. Then give an example showing that the condition cannot be weakened to $|N(S)| \geq |S| - 1$.
3. Let G be a simple graph with n vertices and m edges in which no two cycles share an edge. Prove that $2m < 3n$. (ICMC)
4. In a town every two residents who are not friends have a friend in common, and nobody is a friend of everybody else. Number the residents $1, \dots, n$ and let a_i be the number of friends of the i -th. Suppose $\sum_{i=1}^n a_i^2 = n^2 - n$. Let k be the smallest number of residents, at least three, who can be seated at a round table so that any two neighbours are friends. Determine all possible values of k . (IMC)

5. Prove Euler's formula $v - e + f = 2$ for connected plane graphs, deduce $e \leq 3v - 6$ with equality exactly for triangulations, $e \leq 2v - 4$ in the triangle-free case, and the non-planarity of K_5 and $K_{3,3}$. Deduce also that every planar graph has a vertex of degree at most five (the five-colour theorem that follows from it is proved in Methods of proof – 2, homework 8).